ECA1803 – OPTOELECTRONICS AND OPTICAL COMMUNICATION
CO4 – Investigative Assignment

Ultra-High-Speed Photonic Transport Link: Technology Evaluation, Impairment Mitigation and Integrated System Design

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1. Stage 1 – Interpretation of the Communication Requirement

1.1 Literature and standards reviewed

The link is to be built on the ITU-T 50 GHz DWDM frequency grid (G.694.1) and the amplified multichannel application framework of G.698.2 [1], [2]. IEEE 802.3ct defines 100 Gb/s DWDM transport using DP-DQPSK [3], while IEEE 802.3ck and OIF-CEI-112G define the 100 Gb/s-per-lane PAM4 electrical host interface that feeds the transmitter [4], [5]. Coherent detection with DSP has replaced direct detection above 100 Gb/s per wavelength because it recovers amplitude, phase and polarization and moves dispersion compensation into the digital domain [6], [9]. The nonlinear limit of such links is usually estimated with the Gaussian-noise (GN) model [7], [8], which is used later in this report.

1.2 Aggregate capacity

Aggregate capacity = $N_{ch} \times R_{ch}$ = 32 × 200 Gb/s = **6.4 Tb/s** (net payload).

On a 50 GHz grid the 32 channels occupy 32 × 50 GHz = 1.6 THz, about one third of the 4.8 THz C-band. Net spectral efficiency is therefore 200 Gb/s ÷ 50 GHz = **4 b/s/Hz**.

1.3 Performance metrics to be investigated

Table 1: Key physical-layer metrics and targets

Metric	Target / working value
Spectral efficiency	≥ 4 b/s/Hz (net, 50 GHz grid)
Pre-FEC BER / Q-factor	BER < 3.8×10^{-3} (Q > 8.53 dB)
Required OSNR (0.1 nm)	> 18 dB (theoretical minimum for DP-16QAM is 18.7 dB; calculated in Section 2)
Launch power per channel	≈ 0 dBm (set by ASE–nonlinearity balance, Section 3)
Modulator 3 dB bandwidth	> 35 GHz
Insertion loss (Tx PIC + AWG)	Tx PIC ≈ 17 dB (laser to modulator output, incl. 6 dB QAM loss); AWG ≈ 4.5 dB
Footprint / power	Compact Si-PIC transceiver, ≈ 10 W per 200G channel

1.4 Impairments and the DSP function that handles each

Table 2: Impairment to compensation mapping

Impairment	Nature	Handled by
Chromatic dispersion	Linear, static, large (15,030 ps/nm over 900 km)	Frequency-domain equalizer (FIR/FFT)
PMD, PDL	Linear, time-varying, random	Adaptive 2×2 MIMO equalizer (CMA/RDE)
Laser phase noise, frequency offset	Linear, slow	Frequency-offset estimation, blind phase search
SPM / XPM / FWM	Nonlinear, power dependent	Launch-power optimisation; digital back-propagation (partial)
ASE noise	Additive, cannot be removed	Amplifier design (NF), launch power, FEC
AWG/MMI crosstalk	Linear, interferometric	Component design (≥ 25 dB isolation); MIMO removes part

1.5 Technology trade-offs

Lithium niobate gives the best linearity and bandwidth but is hard to integrate with CMOS electronics and with the receiver. Electro-optic polymers give very low V_π but their long-term thermal and photo-stability is still a reliability risk for a carrier network. Silicon photonics has lower modulation efficiency and higher loss, but it allows the modulator, the coherent receiver, the MMI couplers and the polarization splitter/rotator to sit on one CMOS-compatible chip, which gives the lowest cost, footprint and power per bit [13], [15]. Reliability and thermal stability (heaters, bias control) are therefore weighed against integration level in Stage 2.

Investigation Requirement Statement

The DWDM system shall carry 32 channels × 200 Gb/s = 6.4 Tb/s on the 50 GHz ITU grid (≥ 4 b/s/Hz), over standard SMF-28 with coherent detection and DSP. Each channel shall have a pre-FEC BER below 3.8×10^{-3} (Q > 8.53 dB) with a required OSNR of about 19–21 dB, and at least 2 dB of Q-margin at the end of the link. The transmitter shall use a compact CMOS-compatible photonic modulator with > 35 GHz bandwidth, the amplifiers shall provide ≥ 17 dB gain per span with < 1 dB gain non-flatness, and the receiver DSP shall compensate CD and PMD fully and nonlinearity at least partly, all within G.694.1/G.698.2 and the 802.3ck/OIF-CEI host interface.

2. Stage 2 – Modulation Format and Modulator Selection

2.1 Baud rate and channel spacing

The line rate must be higher than 200 Gb/s because of FEC and framing overhead. A 7 % hard-decision FEC (matching the 3.8×10^{-3} threshold) plus about 5 % for OTN framing and DSP pilot symbols gives 12 % overhead, i.e. a gross rate $R_g = 200 \times 1.12 = 224$ Gb/s. The symbol rate for each format is

$$R_s = R_g / (N_{pol} \cdot log_2 M)$$

DP-16QAM: $R_s = 224 / (2 \times 4) =$ **28 GBd**. With root-raised-cosine shaping (roll-off $\beta = 0.2$) the occupied bandwidth is $R_s(1+\beta) = 33.6$ GHz, so it fits in a 37.5 GHz slot and comfortably in the 50 GHz grid slot used here.

PAM4 (single polarization, intensity): $R_s = 224 / (1 \times 2) =$ **112 GBd**. Direct-detection spectra need about $1.5 \times R_s = 168$ GHz, i.e. a 175 GHz slot. This is 3.5 times wider than the grid slot, and 32 channels would need 5.6 THz, more than the whole C-band. A 112 GBd signal also needs a > 80 GHz modulator and ADC, and its dispersion-limited reach is $L \approx 10^5/(D \cdot B^2) \approx 0.5$ km without DSP, so PAM4 is not suitable at this rate for DWDM transport.

2.2 Comparison of candidate formats

For square M-QAM with Gray coding the pre-FEC BER is $BER \approx [4/\log_2 M](1 - 1/\sqrt{M})\, Q(\sqrt{(3 \cdot SNR/(M-1))})$. Solving for $BER = 3.8\times10^{-3}$ gives the required SNR per symbol, and the required OSNR in 0.1 nm (12.5 GHz) follows from $OSNR = SNR + 10 \log(R_s/12.5\ GHz)$. For the direct-detection formats the OSNR values are approximate, from the Q-OSNR relation with a receiver electrical bandwidth of $0.75\ R_s$.

Table 3: Formats compared for 200 Gb/s net (224 Gb/s gross)

Format	Bits/ symbol	R_s (GBd)	Min. slot (GHz)	SE (b/s/Hz)	Req. OSNR (dB)	Remarks
NRZ-OOK	1	224	337.5	0.59	≈ 19.8	CD-limited to < 1 km; impractical
DPSK	1	224	337.5	0.59	≈ 16.8	Same bandwidth problem as OOK
PAM4	2	112	175	1.14	≈ 26.3	Highest OSNR need of all; used for short reach only
DQPSK (single pol.)	2	112	175	1.14	≈ 20.4	Needs 2 × 56 G interferometer drive
DP-QPSK	4	56	75	2.67	15.0	Long reach, but half the SE
16-QAM (single pol.)	4	56	75	2.67	21.7	No polarization multiplexing

Format	Bits/ symbol	R_s (GBd)	Min. slot (GHz)	SE (b/s/Hz)	Req. OSNR (dB)	Remarks
DP-16QAM	**8**	**28**	**37.5**	**5.33 (4.0 on 50 GHz)**	**18.7**	**Selected**
DP-64QAM	12	18.7	25	8.0	22.9	Needs ≈ 4 dB more OSNR; reach too short

Going from DP-QPSK to DP-16QAM at the same bit rate costs 15.19 − 8.53 = 6.7 dB in SNR (3.7 dB in OSNR, since the baud rate halves), but doubles the spectral efficiency. DP-64QAM would need a further 5.9 dB of SNR, which the amplifier noise and nonlinearity of a multi-span link do not allow. DP-16QAM therefore gives the best balance.

2.3 Modulation mechanisms

The phase shift of a Pockels modulator is $\Delta\varphi = \pi V/V_\pi$ with $V_\pi \cdot L = \lambda g/(n^3\, r\, \Gamma)$, where g is the electrode gap, r the electro-optic coefficient and Γ the overlap factor. For a push-pull Mach-Zehnder the output field is $E_{out} = E_{in} \cos(\pi V/2V_\pi)$. Using λ = 1.55 µm, g = 10 µm and Γ = 0.5 gives $V_\pi L \approx 10.3$ V·cm for bulk $LiNbO_3$ (n = 2.14, r_{33} = 30.8 pm/V), and with g = 5 µm, Γ = 0.8, n = 1.7 and r_{33} = 150 pm/V an EO polymer gives ≈ 1.3 V·cm. For a thermo-optic phase shifter, a π shift needs $\Delta T = \lambda/(2L \cdot dn/dT)$; with silicon $dn/dT = 1.86\times10^{-4}$ K^{-1} and L = 0.5 mm this is 8.3 K.

Table 4: Modulation mechanisms

Mechanism	Speed limit	$V_\pi L$ / drive	Chirp	Use in this link
Electro-optic, plasma dispersion (Si)	30–50 GHz (RC, carrier transit)	≈ 1.5–2.5 V·cm	Moderate, reduced by push-pull	**Data modulation (selected)**
Electro-optic, Pockels ($LiNbO_3$)	> 100 GHz (fs-ps response)	≈ 10 V·cm bulk, 2–3 V·cm thin-film	Near zero (push-pull)	Upgrade path > 56 GBd
EO polymer / SOH	> 100 GHz	0.05–1.3 V·cm	Low	Promising; reliability not yet proven
Thermo-optic	10 µs–ms (kHz)	$dn/dT = 1.86\times10^{-4}$ K^{-1}	None (static)	Bias control and ring tuning only
Acousto-optic	≤ ~1 GHz (acoustic velocity)	RF power, not voltage	Frequency shift	Tunable filters / switches, not data
Magneto-optic (Faraday)	kHz–MHz (coil inductance)	$\theta = V \cdot B \cdot L$	–	Isolators and circulators
Kerr effect	fs, but needs very high field/intensity	No practical voltage	–	An impairment (SPM/XPM), not a modulator

2.4 Mach-Zehnder versus micro-ring structures

Table 5: MZM and MRR in silicon photonics and EO polymer

Parameter	Si travelling-wave MZM	Si micro-ring (MRR)	EO-polymer (SOH) MZM
Bandwidth	35–45 GHz (electrode loss, velocity mismatch)	$f \approx f_0/Q$ = 193.4 THz/5000 ≈ 39 GHz, falls with higher Q	> 100 GHz
Footprint	≈ 5 mm long; ≈ 8 mm^2 for DP-IQ	≈ 10 μm radius, ≈ 100 μm^2	≈ 0.5–1 mm long
Thermal sensitivity	Bias drift, controlled by one bias loop per arm pair	≈ 80 pm/K (≈ 10 GHz/K), needs resonance locking	Polymer glass-transition and drift issues
Insertion loss	4–6 dB	1–3 dB	3–6 dB
Linearity for QAM	Good (sinusoidal, push-pull)	Poor: Lorentzian response, chirp	Good
Suitability for DP-16QAM	Yes (IQ nested MZM)	No single-ring IQ; ring-assisted designs complex	Yes, but reliability risk

DESIGN DECISION 1 – Modulation / Modulator Configuration

I recommend **DP-16QAM at 28 GBd generated by a silicon-photonic dual-polarization IQ travelling-wave Mach-Zehnder modulator (carrier-depletion plasma-dispersion, push-pull)** because it gives 200 Gb/s net in a 37.5 GHz minimum slot (4 b/s/Hz on the 50 GHz grid, 5.33 b/s/Hz on a flexible grid), needs only 18.7 dB theoretical OSNR (≈ 21.0 dB with transceiver noise), and its 28 GBd symbol rate is well inside a modulator of 38 GHz bandwidth. With $V_\pi L$ = 1.8 V·cm and L = 5 mm, V_π = 1.8/0.5 = 3.6 V, so a 3 V_{pp} SiGe driver (≈ 0.83 V_π) is enough. A ring modulator is rejected for its nonlinear response and thermal sensitivity, and polymer devices for reliability. Thin-film $LiNbO_3$ is kept as the upgrade path for 56 GBd and above.

3. Stage 3 – Optical Amplification Configuration

3.1 Amplifier comparison

Table 6: Candidate optical amplifiers

Amplifier	Gain bandwidth	Saturated output	Noise figure	Gain tilt / crosstalk
EDFA	≈ 35 nm (C-band)	+17 to +23 dBm	4.5–6 dB (5.5 dB used)	Strong tilt, needs GFF; no inter-channel crosstalk (slow dynamics)
SOA	40–80 nm	+10 to +15 dBm	7–9 dB	Fast gain dynamics cause XGM and FWM between channels; PDG
Lumped Raman	Set by pumps, up to ≈ 100 nm	n/a (needs 1–2 W pump)	3–5 dB	Flat if multi-pump; high pump power and cost
Distributed Raman (backward)	Set by pumps, tunable	n/a (pre-amplifier)	Effective NF ≈ −1 dB	Flat with 3 pumps; low pump-RIN transfer; laser-safety issues
Hybrid Raman + EDFA	Pump-tuned, 35 nm+	+17 dBm (EDFA stage)	**Equivalent ≈ 0.5 dB**	Raman shapes gain, GFF trims it

3.2 Power budget (analytical problem)

Input per channel = −15 dBm; target output per channel = +2 dBm; N = 32 channels, 10 log 32 = 15.05 dB.

Total input power = −15 + 15.05 = +0.05 dBm ≈ 1.0 mW.

Total output power = +2 + 15.05 = **17.05 dBm = 10^{1.705} mW ≈ 50.7 mW ≈ 0.0507 W**.

Required span gain = $P_{out} - P_{in}$ = 2 − (−15) = **17 dB** per channel (a factor of 50).

The span uses 75 km of SMF-28 at α = 0.2 dB/km, so the fibre loss is 15 dB. The remaining 2 dB is the passive loss at the amplifier site (isolator 0.5 dB, GFF/WSS 1.0 dB, connectors 0.5 dB), which gives a launch power of +2 − 2 = 0 dBm per channel. In the hybrid amplifier the distributed Raman stage provides about 9 dB on-off gain (three backward pumps at 1427, 1450 and 1480 nm, roughly 450 mW in total) and the EDFA provides the remaining 8 dB; the 17 dB total is the same as for an EDFA-only span.

3.3 Gain flatness

The erbium gain spectrum has a peak near 1532 nm and falls towards 1565 nm. In the model the raw EDFA ripple is 6.3 dB peak-to-peak across the C-band, which would cause the weakest channels to lose OSNR directly and the strongest channels to generate more nonlinearity. A thin-film gain-flattening filter, together with pump-power shaping in the Raman stage, reduces the ripple to 0.44 dB per amplifier (Fig. 5). Because residual ripple of the cascade adds roughly

as the root-sum-square, 4 amplifiers give about 0.9 dB, so a WSS-based dynamic gain equalizer is placed after every fourth amplifier to keep $\Delta G < 1.0$ dB along the whole link.

3.4 Cascaded ASE and OSNR

For N identical spans, with the signal power P_{in} at the amplifier input and noise measured in 0.1 nm,

$$OSNR_N\ (dB) = 58 + P_{in} - NF - 10 \log N$$

The constant 58 dB is $-10 \log(h\nu \cdot 12.5\ \text{GHz})$ in dBm at 1550 nm. For the EDFA-only span, $OSNR_1 = 58 - 15 - 5.5 = 37.5$ dB. For the hybrid span the equivalent noise figure is found from the cascade, $F = F_{R,eff} + (F_{EDFA} - 1)/G_R = 0.79 + (3.55 - 1)/7.9 = 1.12$, i.e. $NF_{eq} \approx 0.5$ dB, so $OSNR_1 = 58 - 15 - 0.5 = 42.5$ dB, an improvement of 5 dB.

Table 7: OSNR (dB, 0.1 nm) versus number of 75 km spans

Spans N	1	4	8	12	16	20
Distance (km)	75	300	600	900	1200	1500
EDFA only	37.5	31.5	28.5	26.7	25.5	24.5
Hybrid Raman + EDFA	42.5	36.5	33.5	31.7	30.5	29.5

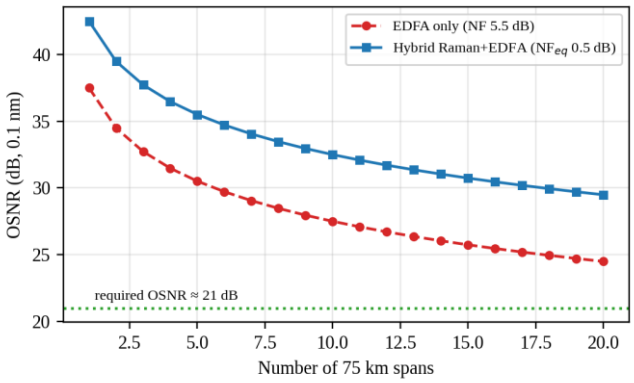


Fig. 1: OSNR versus number of spans for the two amplifier options.

DESIGN DECISION 2 – Amplification Configuration

I recommend a **hybrid amplifier: backward three-pump distributed Raman (≈ 9 dB on-off gain) followed by an EDFA with a GFF (≈ 8 dB), 75 km spans, and a WSS equalizer every fourth span** because the 17 dB span gain and +2 dBm per channel (17.05 dBm total, 50.7 mW) are met, the equivalent noise figure of 0.5 dB gives 5 dB better OSNR than EDFA-only (31.7 dB against 26.7 dB after 12 spans), and the gain flatness stays below 1.0 dB. SOA is rejected because of inter-channel crosstalk and its 7–9 dB noise figure. If the Raman pumps fail the line degrades to EDFA-only operation, which still meets the BER threshold over 900 km (Section 5).

4. Stage 4 – Impairment and DSP Compensation

4.1 Impairment mechanisms and their size in this link

Table 8: Impairments for 28 GBd DP-16QAM over 900 km (12 × 75 km SMF-28)

Impairment	Mechanism	Value for this link
Chromatic dispersion	Group velocity depends on wavelength; $\beta_2 = -D\,\lambda^2/(2\pi c) = -21.3$ ps^2/km	$D \cdot L = 16.7 \times 900 = 15{,}030$ ps/nm. Signal width $\Delta\lambda = 0.224$ nm, so the pulse spreads 3.4 ns ≈ 94 symbols
PMD	Random birefringence gives differential group delay (DGD) ∝ √L	0.1 ps/√km × √900 = 3.0 ps mean DGD (maximum ≈ 10 ps, 0.3 symbol)
PDL	Polarization-dependent loss in components	≤ 0.5 dB per element; budgeted as 1 dB margin
SPM	Intensity-dependent phase, $\varphi_{NL} = \gamma P L_{eff}$	$L_{eff} = 21.0$ km; $\varphi = 1.3 \times 10^{-3} \times 21.0 = 0.027$ rad per span, 0.33 rad over 12 spans at 0 dBm
XPM	Phase modulation by neighbouring channels	Walk-off length ≈ $T_s/(D \cdot \Delta\lambda_{sp}) = 35.7$ ps/(6.7 ps/km) ≈ 5.3 km, which averages XPM; it appears as noise
FWM	Phase-matched mixing of three channels	Efficiency ≈ $\alpha^2/(\alpha^2+\Delta\beta^2)$ ≈ −33 dB at 50 GHz spacing ($\Delta\beta \approx 2.1$ km^{-1}), so negligible

4.2 Optical versus digital compensation

Table 9: Dispersion compensation approaches

Criterion	DCF modules	Optical phase conjugation	Digital (coherent DSP)
Added loss / noise	≈ 6 dB per 75 km span, needs an extra amplifier stage and adds NF	Needs pump and nonlinear medium at mid-span	None (no optical noise added)
Nonlinearity	DCF has high γ, increasing NLI	Compensates SPM only if link is symmetric	Fibre is uncompensated; NLI of the line is lower, DBP optional
Flexibility	Fixed, per fibre type	Fixed location	Adaptive to any distance and fibre
PMD	Not compensated	Not compensated	Compensated adaptively
Cost / scalability	Passive but bulky	Complex, not per channel	Scales with CMOS, one ASIC per channel

An uncompensated line with digital equalization is therefore selected. It removes the DCF modules and one amplifier stage per span, which also improves OSNR by about 1–1.5 dB compared with a DCF-managed link [9].

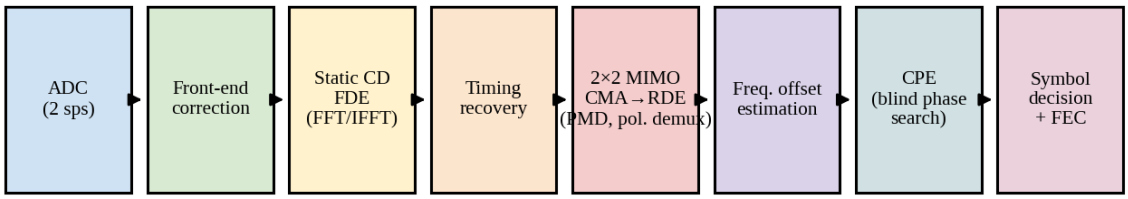
4.3 Coherent DSP blocks

Static CD equalizer. CD is an all-pass filter $H(\omega) = \exp(j \cdot D\lambda^2 L\omega^2/(4\pi c))$. In the time domain a FIR filter with $N_{taps} = 2\lfloor D\,\lambda^2\,L/(2c\,T^2)\rfloor + 1$ is required, where $T = 1/(2R_s) = 17.9$ ps for two samples per symbol. For 900 km this gives $N_{taps} = 377$, which is costly, so the filter is applied in the frequency domain with overlap-save FFT blocks (512-point), reducing the multiplications per sample to about $2\log_2 N + 1 \approx 19$.

Polarization demultiplexing and PMD compensation. A 2×2 butterfly of T/2-spaced FIR filters (15 taps each, covering ≈ 270 ps, far above the 10 ps maximum DGD) is adapted blindly. Since 16-QAM does not have a constant modulus, the filters are first converged with the CMA, $e_k = (R^2 - |y_k|^2)y_k$, and then switched to a radius-directed (RDE) or decision-directed LMS update, $w \leftarrow w + \mu\, e\, x^*$ [20].

Digital back-propagation. DBP solves the inverse nonlinear Schrödinger equation with the split-step Fourier method: each step applies a CD filter of length z/N_s and a nonlinear phase rotation $\exp(-j\gamma|A|^2 L_{eff})$. With one step per span, 12 steps are needed, each requiring an FFT/IFFT pair (≈ 19 + 3 multiplications per sample), i.e. ≈ 12 × 22 ≈ 264 per sample, about 14 times the static CD block. Full-field DBP of all 32 channels would require receiving 1.6 THz coherently and is not realistic. Single-channel DBP only removes SPM and is limited by XPM from the other 31 channels; in the model it gives +0.2 dB in Q at 0 dBm (Fig. 8) [10]. Since the CD block is roughly 10–15 % of the DSP power, 12-step DBP would roughly double the DSP power for this small gain.

4.4 Receiver DSP pipeline



Optional (extended-reach mode): single-channel DBP inserted before the static CD block, replacing it.

Fig. 2: Selected receiver DSP chain.

The order is fixed by the dependencies between blocks. The front-end correction (deskew, IQ imbalance, orthogonalisation) comes first because every later block assumes four balanced, aligned tributaries. CD is removed before timing recovery because the 94-symbol spread destroys the clock tone, and it must also precede the MIMO stage, whose 15 taps cannot cover that memory. The MIMO equalizer follows, undoing the polarization rotation and PMD, and uses the corrected signal to converge. Frequency-offset estimation comes before carrier-phase estimation because the phase estimator can only track small residual offsets. Blind phase search with 32 test phases is used for 16-QAM, since the laser linewidths (100 kHz at Tx and LO)

give $\Delta\nu \cdot T_s$ = 200 kHz/28 GBd ≈ 7×10^{-6}, well below the 10^{-4} that 16-QAM tolerates with less than 1 dB penalty [21].

DESIGN DECISION 3 – DSP Compensation Configuration
I recommend **an uncompensated line with a fully digital receiver: front-end correction → frequency-domain CD equalizer (512-point FFT, 377-tap equivalent) → timing recovery → 2×2 MIMO CMA/RDE equalizer (15 taps, T/2) → frequency-offset estimation → blind-phase-search CPE → decision/FEC, with single-channel DBP kept only as an optional mode for links longer than 1,000 km** because this chain compensates all of the 15,030 ps/nm of CD and the PMD, needs no DCF, and gives Q = 10.56 dB (BER 3.7×10^{-4}) after 900 km, which is 2.0 dB above the 8.53 dB threshold. Single-channel DBP adds only 0.2 dB in Q (BER 2.8×10^{-4}) for about twice the DSP power, so it is not justified in the baseline design.

5. Stage 5 – Integrated WDM Transmission Strategy

5.1 Global performance

The link performance was calculated with the GN model [8] for 32 channels at 50 GHz spacing (γ = 1.3 $W^{-1}km^{-1}$, D = 16.7 ps/nm/km, coherence factor ε = 0.05), with the transceiver back-to-back SNR limited to 19.0 dB. The total SNR is $1/SNR = 1/SNR_{ASE} + 1/SNR_{NLI} + 1/SNR_{TRx}$. At the optimum launch power of about 0 dBm per channel, after 12 spans: SNR_{ASE} = 28.2 dB, SNR_{NLI} = 23.6 dB and the total SNR is 17.3 dB for the hybrid line. The corresponding Q-factors are in Table 10, and the GN-model Q-factor against launch power is in Fig. 8.

Table 10: Link performance after 12 × 75 km (900 km), launch 0 dBm per channel

Configuration	OSNR (dB)	SNR (dB)	Q (dB)	BER	Margin over 8.53 dB
EDFA only	26.7	16.6	9.88	9.1×10^{-4}	1.35 dB
Hybrid Raman + EDFA	31.7	17.3	10.56	3.7×10^{-4}	2.0 dB
Hybrid + single-channel DBP	31.7	17.5	10.76	2.8×10^{-4}	2.2 dB

The OSNR needed at the receiver, including the transceiver noise, follows from the 15.19 dB SNR threshold: $1/SNR_{line} = 1/33.0 - 1/79.4$, so SNR_{line} = 17.5 dB and $OSNR_{req}$ = 17.5 + 3.5 = 21.0 dB. The hybrid link therefore has an OSNR margin of 31.7 − 21.0 = 10.7 dB, although the usable margin is set by the transceiver and nonlinear noise (2.0 dB in Q). For comparison, the Shannon limit at SNR = 17.3 dB is $2 \times 28 \times \log_2(1 + 53.7) \approx 324$ Gb/s, above the 224 Gb/s gross rate, so a margin for the FEC and DSP exists.

Estimated power: a 200G coherent transceiver needs about 10 W (DSP with ADC/DAC 5.5 W, driver and TIA 1.5 W, two lasers 2.0 W, thermal control 1.0 W) = 50 pJ/bit. The line needs about 17 W per amplifier site (three Raman pump modules 7.5 W, EDFA 6 W, control 3.5 W) × 12 = 200 W, plus 20 W for booster and pre-amplifier. In total 32 × 10 + 200 + 20 = 540 W, i.e. ≈ **84 pJ/bit** for 6.4 Tb/s. The silicon chip area is about 8 mm × 4 mm per transceiver, ≈ 10 cm^2 for 32 channels.

5.2 Inter-dependencies

Modulation order and OSNR/DSP. Moving from QPSK to 16QAM raises the SNR requirement by 6.7 dB, which directly limits the number of spans (OSNR falls by 10 log N), tightens the transceiver SNR requirement (≥ 19 dB means an ADC/DAC ENOB of about 5 bits), makes the laser-linewidth tolerance about four times stricter and requires RDE and a 32-phase BPS instead of the simpler CMA and 4th-power estimator used for QPSK.

MMI/AWG loss and launch power. The launch budget from the transmitter is: laser +16 dBm, fibre coupling −1.5 dB, X/Y split in the MMI tree −3.5 dB, nested IQ modulator −5.0 dB, average 16-QAM modulation loss −6.0 dB, polarization combiner −1.0 dB, giving −1.0 dBm per channel; the AWG multiplexer adds −4.5 dB, so −5.5 dBm per channel enters the booster. The booster has to give +7.5 dB to reach +2 dBm per channel (total 17.05 dBm, within a +20 dBm saturated output). Every extra decibel of MMI or AWG loss is either taken from the booster headroom or lost in OSNR, and since NLI grows as P^3, loss at the transmitter should be recovered in the booster and not by raising the launch power.

5.3 Trade-offs: reach, density and feasibility

Table 11: Trade-offs between alternative designs

Option	Channels in C-band	Capacity	Required OSNR	Reach	Comment
DP-QPSK, 56 GBd, 75 GHz	64	12.8 Tb/s	15.0 dB	Long (≈ 3×)	Longer reach, lower SE
DP-16QAM, 28 GBd, 50 GHz	**96**	**19.2 Tb/s**	**18.7 dB**	**900 km (this design)**	**Balanced**
DP-16QAM, 28 GBd, 37.5 GHz	128	25.6 Tb/s	18.7 dB + filtering penalty	Shorter	Needs sharper filters, WSS
DP-64QAM, 18.7 GBd, 25 GHz	192	38.4 Tb/s	22.9 dB	Short (< 300 km)	Metro only

For metro and regional distances up to about 1,000 km, DP-16QAM gives the highest spectral efficiency that the OSNR budget supports. For more than 1,000 km, the same hardware can switch to DP-QPSK or probabilistically shaped constellations. In the model the hybrid line

reaches about 2,400 km, the EDFA-only line about 1,100 km and the hybrid line with single-channel DBP about 2,600 km while keeping 1 dB of margin; these are ideal-case values and a field reach of about 70 % of them is a safer estimate because of ageing, PDL and ROADM filtering.

5.4 Failure points, scalability and standards

- **Transceiver SNR floor:** at 19 dB it limits the whole link; any degradation of drivers or ADC linearity directly eats the 2 dB margin.
- **Thermal and bias drift of Si MZMs:** requires closed-loop bias control; laser wavelength drift beyond ±2.5 GHz would cut into the 50 GHz AWG passband.
- **Raman pump failure and safety:** the line falls back to EDFA-only (Q = 9.88 dB, still above threshold); automatic laser shutdown according to IEC 60825-1 is needed because of the high pump power on the fibre.
- **PDL and filter narrowing in ROADMs:** both accumulate and are only partly corrected by the MIMO stage; they are covered by the 1 dB budget.
- **Scalability:** 96 channels fit in the C-band (19.2 Tb/s), 128 with 37.5 GHz slots; the L-band doubles this.
- **Standards:** the 50 GHz grid follows G.694.1 and the amplified-line concept follows G.698.2. IEEE 802.3ct specifies 100 Gb/s DP-DQPSK, so the 200 Gb/s DP-16QAM line side proposed here is an extension (as is OIF 400ZR for 400 Gb/s) and not a direct 802.3ct class; the host electrical interface uses 802.3ck / OIF-CEI-112G PAM4 lanes, so it is compliant on that side.

DESIGN DECISION 4 – Integrated WDM Strategy

I recommend **adopting the integrated 32 × 200 Gb/s system (Si-photonic DP-16QAM transmitters, AWG mux/demux on the 50 GHz grid, 12 × 75 km hybrid Raman-EDFA spans, and coherent receivers with the CD/MIMO/BPS DSP) for 900 km metro/regional deployment** because the cross-stage results are consistent: 6.4 Tb/s at 4 b/s/Hz, OSNR 31.7 dB against 21.0 dB required, Q = 10.56 dB against 8.53 dB (2.0 dB margin), $\Delta G < 1$ dB, and about 84 pJ/bit. The critical risk is the 19 dB transceiver SNR, which is handled by the margin and by the fall-back to EDFA-only operation that still passes.

6. Stage 6 – Verification and Final System Design

6.1 Verification table

Table 12: Target versus achieved values

Requirement / parameter	Specified target	Achieved / evidence
Aggregate WDM capacity	6.4 Tb/s (32 × 200 Gb/s)	6.4 Tb/s net (7.17 Tb/s gross line rate)
Spectral efficiency and modulation	≥ 4 b/s/Hz (DP-16QAM)	4.0 b/s/Hz on 50 GHz grid (5.33 on 37.5 GHz); DP-16QAM, 28 GBd
Modulator bandwidth and footprint	> 35 GHz, compact PIC	38 GHz (3 dB), V_π = 3.6 V, ≈ 8 mm^2 per DP-IQ modulator
Amplifier gain and flatness	≥ 17 dB/channel, ΔG < 1.0 dB	17 dB (9 dB Raman + 8 dB EDFA); 0.44 dB per amplifier, ≈ 0.9 dB with WSS every 4 spans
OSNR / BER	OSNR > 18 dB, BER < 3.8×10^{-3}	OSNR 31.7 dB (required 21.0 dB); BER 3.7×10^{-4}, Q = 10.56 dB
DSP impairment mitigation	Full CD/PMD and nonlinear tolerance	CD 15,030 ps/nm fully compensated; DGD ≥ 100 ps tolerated; nonlinearity only partly (+0.2 dB Q with single-channel DBP; optimum launch 0 dBm)
Standards and feasibility	ITU-T G.698 and IEEE 802.3ck	50 GHz grid (G.694.1/G.698.2) and 802.3ck host interface compliant; line side is an extension of 802.3ct

6.2 System block diagram

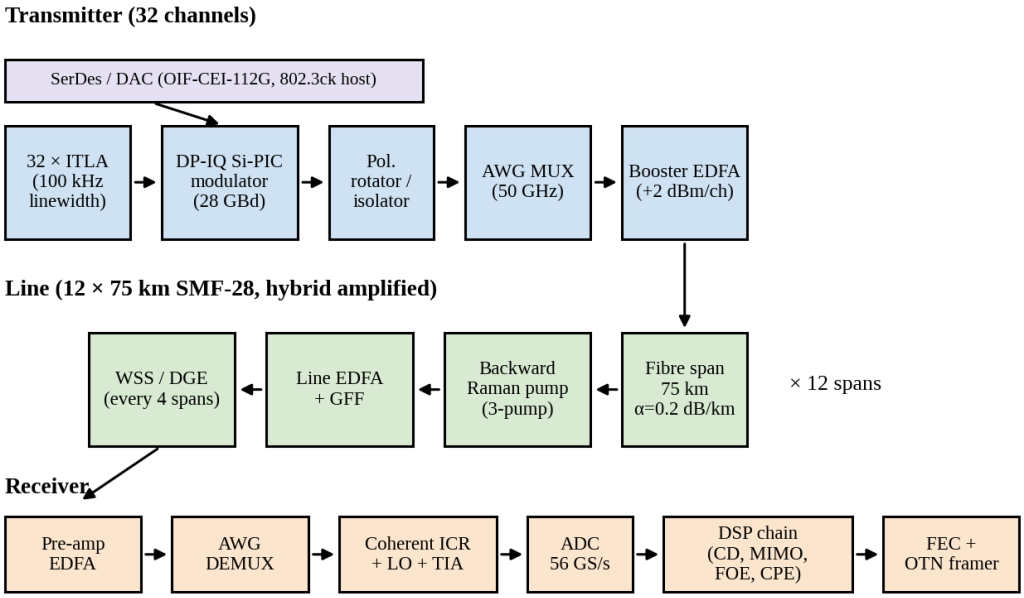


Fig. 3: Block-level architecture of the 32-channel DWDM link.

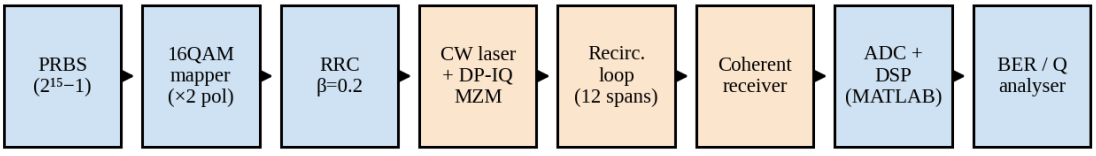
6.3 System design parameters

Table 13: Complete design parameter table

Parameter	Value
Carrier frequencies	192.35 to 193.90 THz (1558.6 to 1546.1 nm), 32 channels
Grid spacing	50 GHz (G.694.1), anchor 193.1 THz
Laser linewidth (Tx and LO)	100 kHz (ITLA), output +16 dBm
Modulation / symbol rate	DP-16QAM, 28 GBd, RRC roll-off 0.2; gross 224 Gb/s
Modulator	Si carrier-depletion IQ-MZM, $V_\pi L$ = 1.8 V·cm, L = 5 mm, V_π = 3.6 V, drive 3 V_{pp}, BW 38 GHz
Fibre	SMF-28: α = 0.2 dB/km, D = 16.7 ps/nm/km, γ = 1.3 $W^{-1}km^{-1}$, PMD 0.1 ps/√km, L_{eff} = 21 km
Span / link	75 km (15 dB loss) × 12 spans = 900 km
Launch power	0 dBm/channel (+2 dBm amplifier output, 2 dB passive loss); 17.05 dBm total at amplifier output
Amplifier	Backward Raman 9 dB (NF_{eff} = −1 dB) + EDFA 8 dB (NF = 5.5 dB); NF_{eq} = 0.5 dB; GFF, WSS every 4 spans
Multiplexer / demultiplexer	AWG, 50 GHz, ≈ 4.5 dB insertion loss, adjacent-channel isolation ≥ 25 dB
Receiver	Coherent ICR, ADC at 56 GS/s (2 sps), ≈ 5-bit ENOB, SNR_{TRx} = 19 dB
DSP taps	CD: 512-pt FFT (377-tap equivalent); MIMO: 2×2, 15 taps T/2; BPS: 32 test phases; FEC: 7 % HD, threshold 3.8×10^{-3}

6.4 Simulation results

The system was modelled as shown in Fig. 4: a random 16-QAM sequence on both polarizations is pulse shaped, modulated, propagated through the recirculating 75 km hybrid span and detected coherently, followed by the DSP chain of Fig. 2.



Shaded blocks map to OptiSystem components (IQ/Mach-Zehnder modulator, Optical Fibre, EDFA, Raman, Coherent Receiver).

Fibre: 75 km SMF-28, D = 16.7 ps/nm/km, γ = 1.3 /W/km, PMD = 0.1 ps/√km; EDFA NF = 5.5 dB; Raman effective NF = −1 dB.

Fig. 4: Simulation model layout.

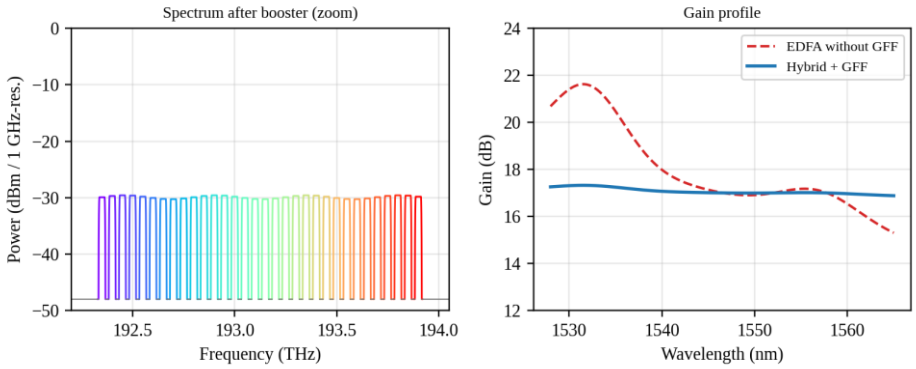


Fig. 5: Left: 32-channel optical spectrum after the booster. Right: amplifier gain profile with and without gain flattening.

The spectrum shows 32 equal-power channels on the 50 GHz grid with a power variation below ±0.4 dB. The gain profile shows the reduction of the ripple from 6.3 dB to 0.44 dB.

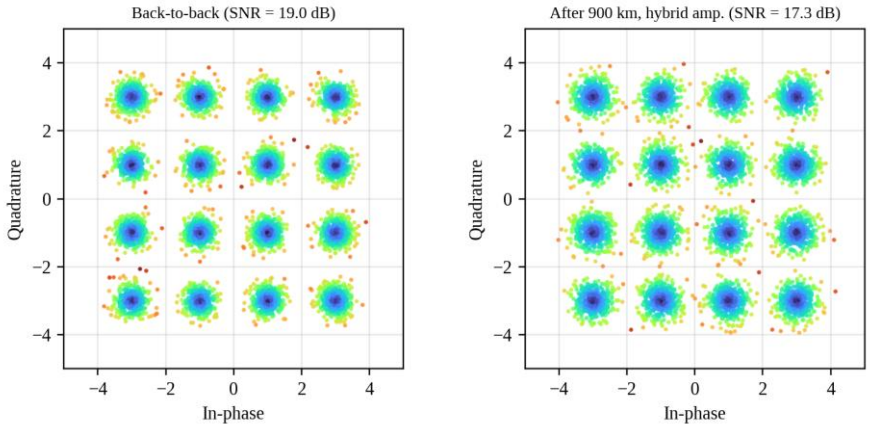


Fig. 6: Received 16-QAM constellation (one polarization), back-to-back and after 900 km.

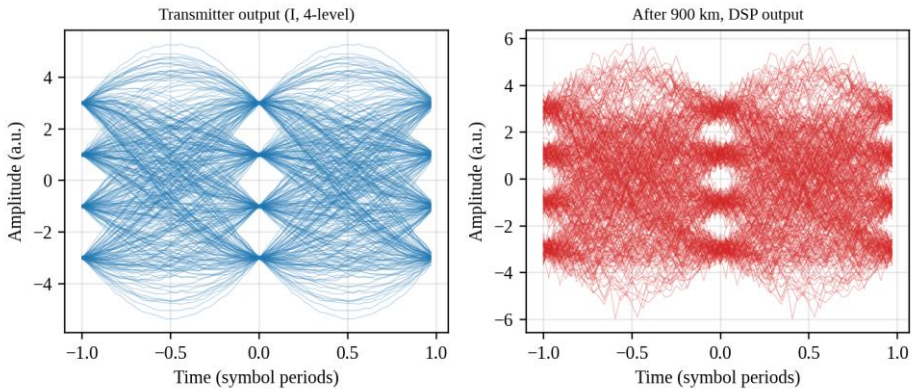


Fig. 7: Four-level eye diagram of the I component at the transmitter and after DSP at 900 km.

After 900 km (SNR 17.3 dB) the 16 clusters are still separated, with occasional overlap of neighbouring symbols consistent with the BER of 3.7×10^{-4}, and the four-level eye remains open.

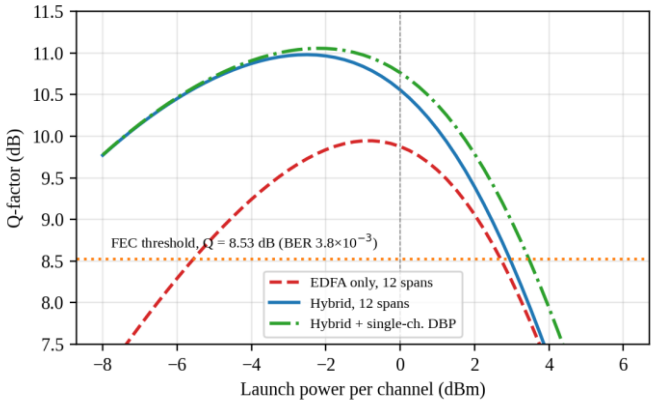


Fig. 8: Q-factor versus launch power per channel after 12 spans.

The curves show the usual maximum. For the EDFA-only line the optimum is near −0.5 dBm and for the hybrid line near −2 dBm, because the lower ASE moves the balance towards lower nonlinear noise; the chosen 0 dBm lies within 0.3 dB of the optimum for the hybrid line. The curves lie 1.4–2.2 dB above the threshold line over a launch range of about ±3 dB, so the operating point is tolerant to power errors.

7. Final Engineering Decision

WOULD YOU DEPLOY THE INVESTIGATED AND DESIGNED WDM SYSTEM? – YES

The system is deployable for 900 km metro/regional transport. The justification, point by point, is:

- **Modulation format efficiency and OSNR sensitivity:** DP-16QAM at 28 GBd gives 4 b/s/Hz on the 50 GHz grid and needs 18.7 dB theoretical (21.0 dB practical) OSNR; the link delivers 31.7 dB.
- **Modulator integration, bandwidth and drive:** the silicon IQ-MZM has 38 GHz bandwidth (> 35 GHz), V_π = 3.6 V driven by a 3 V_{pp} SiGe driver, and is integrated with the MMI tree and polarization combiner on one chip of ≈ 8 mm^2.
- **Amplification, noise and power budget:** 17 dB per span (17.05 dBm total output), NF_{eq} = 0.5 dB, 5 dB OSNR gain over EDFA-only, ΔG ≈ 0.9 dB, and graceful fall-back to EDFA-only.
- **DSP effectiveness:** CD (15,030 ps/nm) and PMD are compensated completely; the nonlinear penalty is controlled by the 0 dBm launch power, with Q = 10.56 dB (2.0 dB margin); DBP is optional because its 0.2 dB gain costs about twice the DSP power.

- **Power, scalability and cost:** ≈ 84 pJ/bit, ≈ 540 W for 6.4 Tb/s, upgradeable to 96–128 channels and to thin-film $LiNbO_3$ or higher baud rates, using CMOS-compatible silicon photonics.

The conditions are that the transceiver SNR stays at or above 19 dB, that bias and wavelength control are implemented, and that the field margin is verified on the real fibre PMD and PDL. Because the answer is YES, no redesign is proposed; for routes longer than 1,000 km the same hardware would use DP-QPSK at 56 GBd on 75 GHz (OSNR requirement 15.0 dB) with optional single-channel DBP.

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